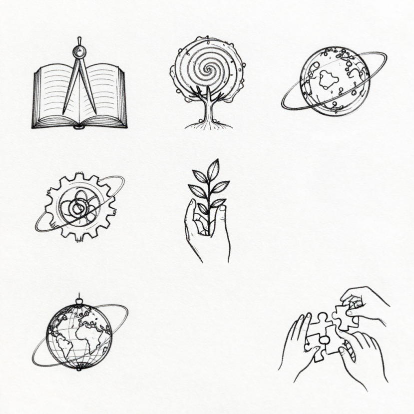

Technocosmos and Frontier

Contents

Why Expansion

Short Answer

Expansion is needed not for a “beautiful race,” but for the long-term survival and maturation of civilization.

Three Rational Reasons

1. **Risk of Concentration on One Planet**
   If all infrastructure, knowledge, and production are tied to Earth, any major global failure becomes existential.
2. **Expanding the Horizon of Science**
   Many questions about life, matter, and sustainability cannot be fully resolved without going beyond Earth.
3. **New Contour of Creative Labor**
   Space projects require long-term cooperation, discipline, and an engineering culture that also strengthens earthly institutions.

What Expansion Should NOT Mean

- not an escape from earthly problems;
- not a justification for social injustice;
- not a cult of superiority of some people over others.

Criterion for Mature Expansion

A space project is justified if simultaneously:

- it gives measurable benefit to life on Earth;
- it does not worsen the basic conditions for people “here and now”;
- it develops a culture of knowledge, rather than just symbolic prestige.

Earth as Rear and Shipyard

Double Role of the Earth

Earth is simultaneously:

- **home**: a place where people, families, culture, and memory live;
- **shipyard**: a base for preparing knowledge, personnel, and technology for distant projects.

Why this is Important

When Earth is perceived only as a "resource," the human foundation of civilization is destroyed.
When Earth is perceived only as a "museum," the horizon of development is lost.
The Code holds both vectors in one contour.

Practical Balance

1. Earthly ecosystems and human safety are a non-negotiable priority.
2. Scientific and engineering preparation for space tasks is a long-term priority.
3. Every major space project must have a "return channel of benefit" for Earth.

Examples of the Return Channel of Benefit

- new materials and medical technologies;
- accurate monitoring of climate and natural risks;
- improving energy efficiency and reliability of infrastructure.

Short Conclusion

Earth is not a “previous version” that needs to be abandoned.
It is the living foundation from which future expansion grows.

Frontier Professions

Why this is not a “Fantasy about the Future”

Frontier professions are already forming: names are changing, but the essence is the management of complex systems over a long distance and in high-risk conditions.

Key Roles

- **Closed Systems Engineer**
  Designs contours of water, air, energy, and waste where an error equals a threat to life.
- **Interorbital Logistics Dispatcher**
  Manages routes, launch windows, and resources under time and fuel constraints.
- **Robotic Construction Operator**
  Deploys infrastructure in hard-to-reach environments with minimal human participation in the risk zone.
- **Habitat Bioengineer**
  Responsible for the sustainability of life contours and biosafety in a long horizon.
- **Failure and Recovery Specialist**
  Does not “fix after an accident,” but builds an architecture where a failure does not turn into a catastrophe.

What Unites these Professions

1. High price of error.

2. Need for team trust.
3. Connection with benefit for Earth (technologies, methods, reliability standards).

Social Meaning

These professions are not about elite status, but about serving the sustainability of civilization.
Their value is measured not by romance, but by the quality of preserved life and created opportunities.

Ethics of Risk and Responsibility

Basic Principle

Risk is permissible if it is conscious, proportional to the goal, and not hidden behind beautiful slogans.

What Makes Risk Ethical

1. People know the real conditions and consequences.
2. There is a right to informed consent and refusal.
3. The system is prepared in advance for failures and accidents.
4. Responsibility is distributed and publicly fixed.

What Makes Risk Unethical

- hiding data about probable losses;
- replacing preparation with heroic rhetoric;
- shifting damage to those who do not make decisions;
- lack of mechanisms for analyzing errors and compensating for harm.

Price of Error

In complex projects, an error is inevitable.
What is unethical is not the error itself, but a culture where:

- it is hidden;
- the guilty are assigned instead of analyzing the causes;
- lessons do not turn into new standards.

Practical Criterion

If after an incident the system has become more transparent, safer, and more honest — the risk was handled in a mature way.
If after an incident fear, lies, and the search for "scapegoats" have grown — this is a failure of responsibility.

Synthesis with the World of Care

False Conflict

Usually contrasted:

- "either we develop technologies and go into space,"
- "or we preserve humanity, closeness, and care."

In the Code, this is a false choice. Without care, technologies become a cold tool, and without development, care loses resource and scale.

Principle of Synthesis

Any large project is evaluated by a double criterion:

1. does it strengthen the ability of people to protect life and relationships;
2. does it expand the horizon of knowledge and opportunities.

If only one point is fulfilled, the project is considered incomplete.

Where the “World of Care” Manifests

- in respect for family and close ties;
- in the ethics of education and upbringing;
- in the culture of helping the vulnerable;
- in honest journalism and open knowledge.

Where “Technocosmos” Manifests

- in complex engineering;
- in scientific discipline;
- in a culture of risk with responsibility;
- in long-term projects that go beyond one generation.

Short Conclusion

A mature civilization does not choose between the warm human and the complex technological.
It builds conditions where one strengthens the other.